

2018 Spring VLSI DSP Homework Assignment #1

Due date: 2018/3/20

1. For a linear system $\mathbf{A}_{4 \times 4} \cdot \mathbf{x}_{4 \times 1} = \mathbf{b}_{4 \times 1}$

$$\mathbf{A}_{4 \times 4} = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & -1 & -1 & -2 \\ 3 & 1 & 0 & 4 \\ 1 & -1 & 5 & 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 11 \\ -13 \\ 21 \\ 14 \end{bmatrix}$$

Please convert $\mathbf{A}$ to an upper triangular matrix using QR decomposition and then use backward substitution to solve the vector $\mathbf{x}$. The QR decomposition process can be described by the following equation

$$\mathbf{Q}_6 \cdots \mathbf{Q}_2 \cdot \mathbf{Q}_1 \cdot \mathbf{A}_{4 \times 4} = \mathbf{R}_{4 \times 4}, \text{ where}$$

$\mathbf{Q}_i, i = 1 \sim 6$ each is a Givens rotation

- write a Matlab code that generates $\mathbf{Q}_i, i = 1 \sim 6$ automatically.
- Compute the equivalent matrix $\tilde{\mathbf{Q}} = \mathbf{Q}_6 \cdots \mathbf{Q}_2 \cdot \mathbf{Q}_1$ and verify it is also a unitary triangular matrix, i.e., $|\det(\tilde{\mathbf{Q}})| = 1$ and $\mathbf{Q} = (\tilde{\mathbf{Q}})^{-1} = \text{transpose}(\tilde{\mathbf{Q}})$
- show that $\mathbf{A} = \mathbf{Q} \cdot \mathbf{R}$ (QR decomposition)
- use the QR decomposition function supported in MatLab, and compare the if the result is identical to what you get
Note: the QR decomposition function in matlab is `qr()`, which returns two matrices, i.e. `[Q,R]=qr(A)`
- Is QR decomposition unique (except for a sign change, i.e. Q,R versus -Q,-R)?

2. Least square optimization problem

For an over constrained linear system, find the least square solution of $\mathbf{x}$ using

a) Pseudo inverse $\hat{\mathbf{x}} = \mathbf{A}^+ \cdot \mathbf{b}$

b) QR decomposition,

c) Compare if a) and b) yield the same result?

Note: the pseudo inverse function is `pinv()` in matlab

$$\mathbf{A} \cdot \mathbf{x} = \mathbf{b}$$

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 3 & 6 \\ 2 & 0 & 5 & 1 \\ 3 & -2 & 1 & 4 \\ -2 & 1 & -1 & -1 \\ 1 & 2 & 4 & 0 \\ 0 & -1 & -2 & 5 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ 1 \\ 3 \\ -2 \\ 4 \end{bmatrix}$$